GradientCIO

PORTFOLIO ATTRIBUTION COMPARISON

Northstar Attribution Comparison

Historical, forward and simulated views

PORTFOLIO
Northstar Total Fund

VIEWS
Three

PREPARED FOR
Investment Committee

DATA AS OF
2026-10-08

01 Executive Comparison

Three compatible governed lanes have server-returned comparison metrics

Three compatible attribution views compared

BOTTOM LINE

Historical, ex ante, and simulated effects are compatible, and governed comparison metrics are available S2 S3 S4 S5.

HISTORICAL LANE
Available
Realized 12 months · 2025-07-01 to 2026-06-30 S2

EX ANTE LANE
Available
1-year expected S3

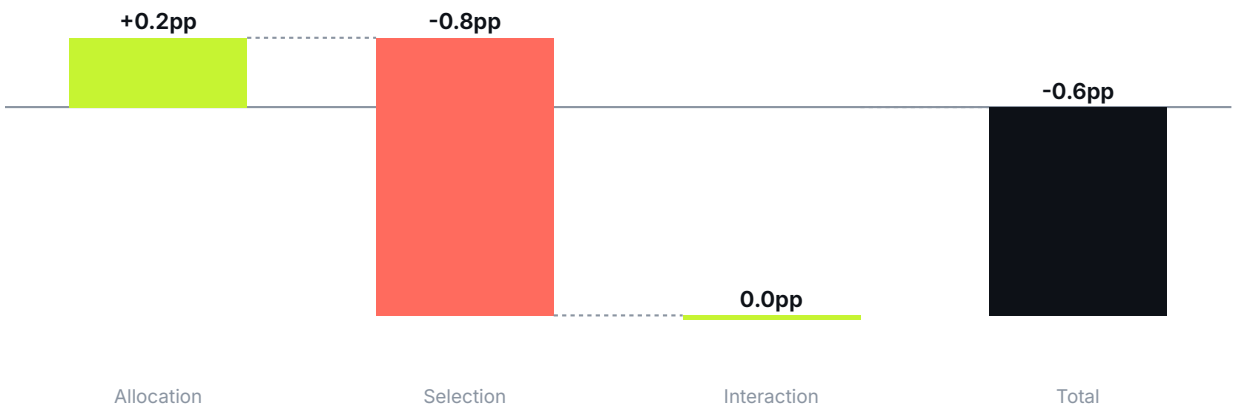
SIMULATED LANE
Available
1-year simulated · governed scenario S4

COMPARISON COVERAGE
Complete
Server-returned comparison · compatible receipts S5

Three governed returned lanes and comparison metrics are available S2 S3 S4 S5.

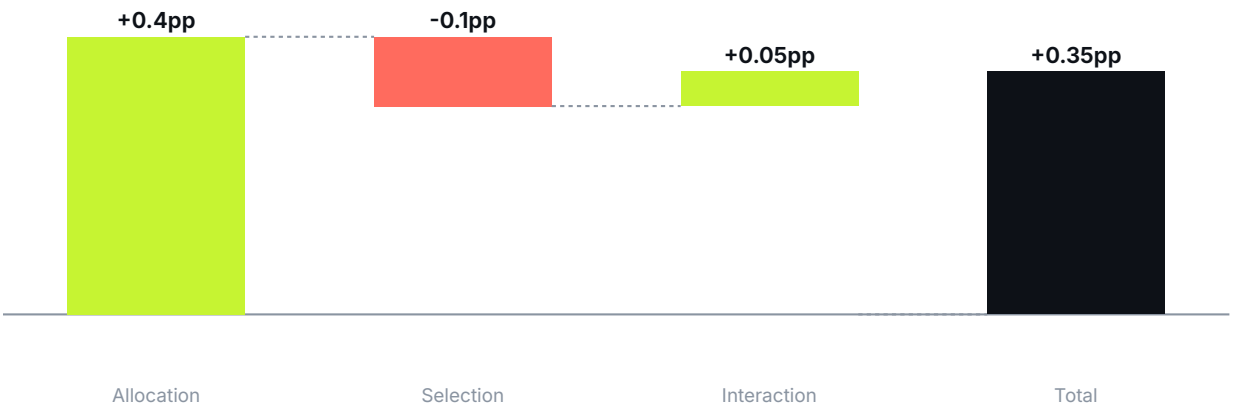
Returned evidence only; no local subtraction or accumulation.

REALIZED SUMMARY EFFECTS



Each plotted value is a returned summary field S2.

EXPECTED SUMMARY EFFECTS



Each plotted value is a returned summary field S3.

Key judgment — Historical

Observation: Historical effects are returned S2. Why it matters: Realized sources can be described. Uncertainty: One period may not persist. What would change the view: A later governed period.

Key judgment — Ex ante

Observation: Ex ante effects are returned S3. Why it matters: Expected sources can be described. Uncertainty: Inputs are assumptions. What would change the view: Refreshed governed inputs.

Key judgment — Comparison

Observation: Server-returned comparison metrics cover three compatible lanes S5. Why it matters: Cross-lane evidence is governed. Uncertainty: One comparison receipt limits time-series interpretation. What would change the view: A later receipt returning different lane signs.

02 Scope and Compatibility

Portfolio, benchmark, cohort and currency identifiers match

Compatible for governed three-lane comparison

The returned compatibility receipt matches portfolio, benchmark role, parent cohort and calculation currency S5.

Compatibility gate	**Available**	Four identifiers match S2 S3

03 Historical Attribution

Returned realized effects are shown without recalculation

HISTORICAL EFFECTS BY SEGMENT



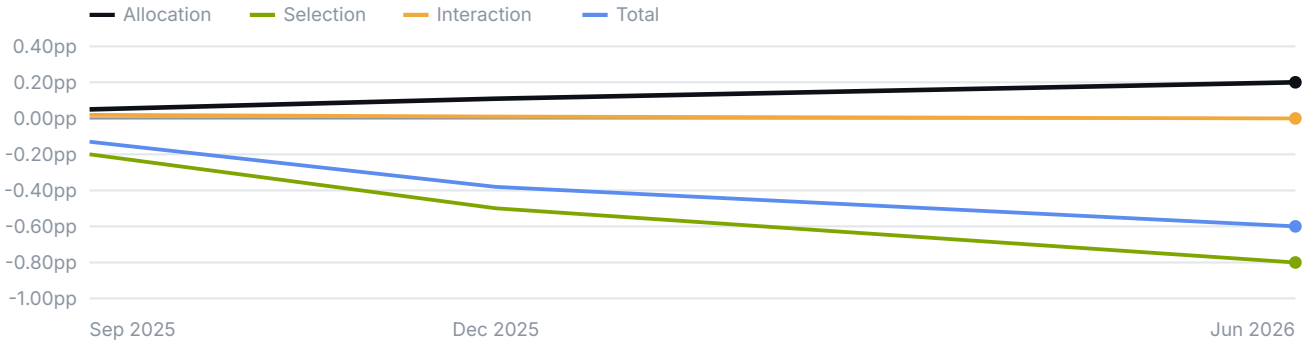
Returned allocation, selection and interaction effects by segment S2.

HISTORICAL PORTFOLIO AND BENCHMARK AVERAGE WEIGHTS



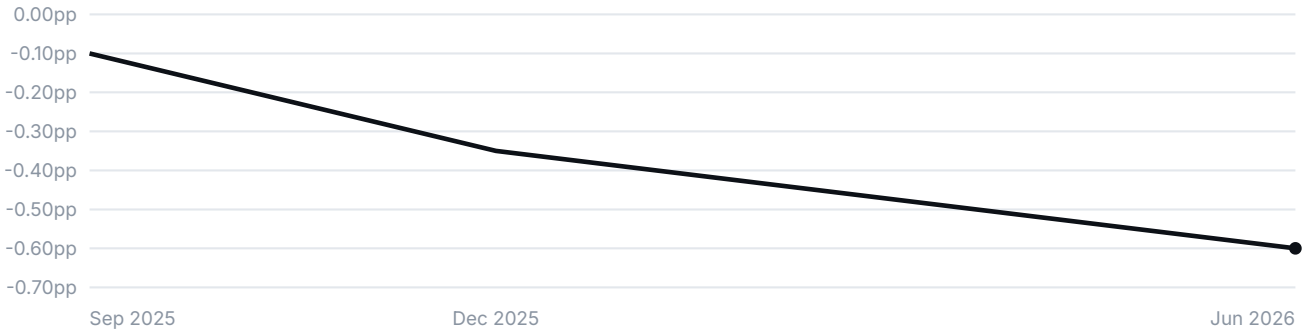
Returned arithmetic average beginning weights; no local difference S2.

CUMULATIVE LINKED EFFECTS



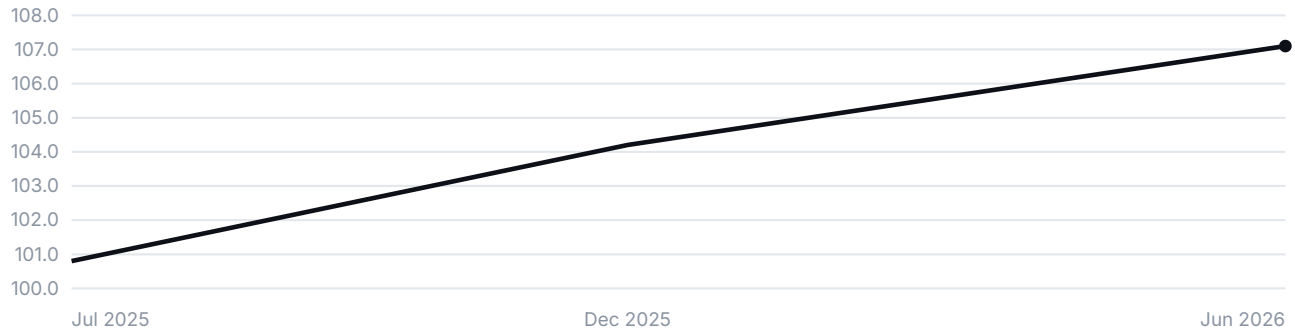
Server-returned cumulative linked fields; no local accumulation S2.

CUMULATIVE ALTERNATIVES SELECTION



Returned rows filtered to allocation_name=Alternatives; no local accumulation S2.

RETURNED CUMULATIVE GROWTH CONTEXT



Optional context rendered only from returned cumulative_growth rows S6.

Analysis — Selection leads detractors

Observation: Returned selection is negative S2. Why it matters: Selection leads realized detractors. Uncertainty: The evidence covers one linked period. What would change the view: A later historical result.

Benchmark and excess boundary

Attribution excess is measured against the linked segment benchmark, not the policy benchmark used for historical returns, so the two excess figures are not directly comparable S2.

04 Governed Ex Ante Attribution

Returned expected effects are shown as assumptions

EXPECTED EFFECTS BY SEGMENT

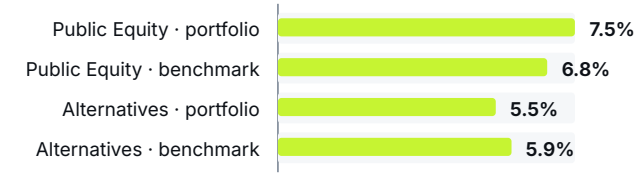


Returned expected allocation, selection and interaction by segment S3.

EX ANTE ASSUMPTIONS

Assumption set	**2026 Strategic CMA** S3
Regime	**User-defined regime** S3
Horizon	**1 year** S3
Portfolio return basis	**Base-case expected return by allocation** S3
Benchmark return basis	**Base-case expected return by benchmark segment** S3

EXPECTED PORTFOLIO AND BENCHMARK RETURNS



Returned expected returns grouped without local subtraction S3.

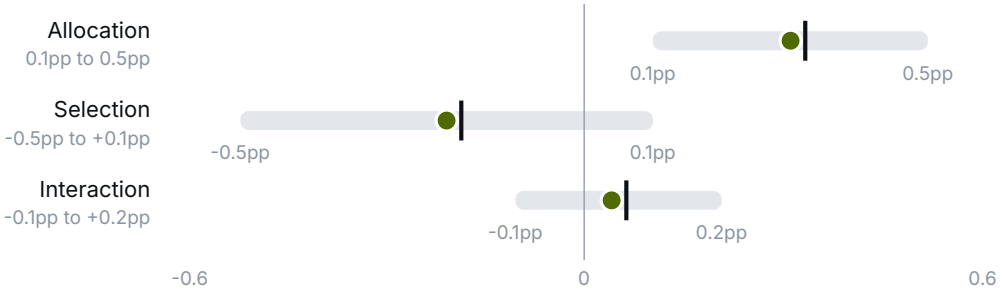
Analysis — Allocation leads expectations

Observation: Returned allocation is positive S3. Why it matters: Allocation leads expected effects. Uncertainty: Inputs are assumptions. What would change the view: A refreshed ex ante result.

05 Simulated Attribution

Governed simulation returns scenario effects

SIMULATED EFFECT BANDS



Server-returned scenario bands and selected values S4.

SCENARIO SELECTOR

SCENARIO ID	SCENARIO	SELECTED	BASIS
scenario-base-test	Base scenario	Yes	1-year simulated
scenario-down-test	Downside scenario	No	1-year simulated
scenario-up-test	Upside scenario	No	1-year simulated

Returned scenario selector; no locally invented scenarios S4.

Capability gate — **Available** Exact simulation capability was available before invocation S1 S4

Analysis — Scenario range returned

Observation: The governed simulation returned effect bands and a selected base scenario S4. Why it matters: Scenario context is explicit. Uncertainty: Results depend on governed scenario inputs. What would change the view: A revised simulation receipt.

06 Cross-Lane Comparison

Governed comparison returns compatible cross-lane metrics

RETURNED CROSS-LANE EFFECT MATRIX

	REALIZED	EXPECTED	SIMULATED
ALLOCATION	0.20	0.40	0.30
SELECTION	-0.80	-0.10	-0.20
INTERACTION	0.00	0.05	0.04

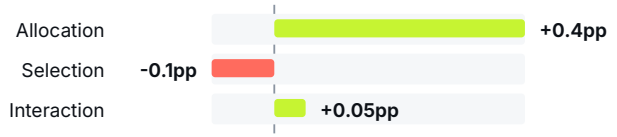
Server-returned three-lane effect matrix S5.

REALIZED RETURNED EFFECTS



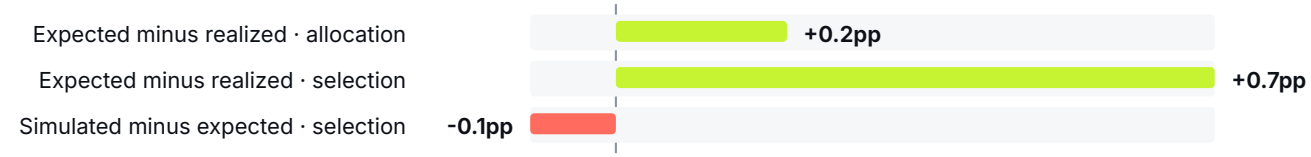
Displayed unchanged S2.

EXPECTED RETURNED EFFECTS



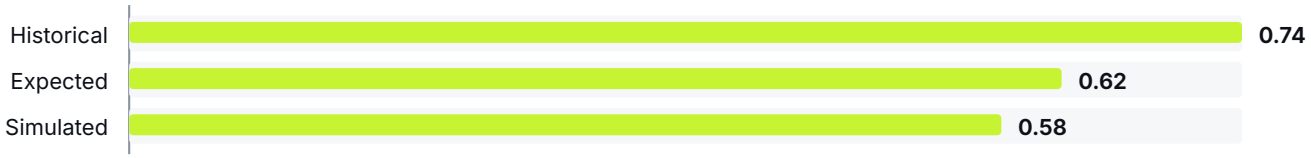
Displayed unchanged S3.

RETURNED LANE DIFFERENCES



Differences are returned by the comparison tool, not calculated here S5.

RETURNED CONCENTRATION



Server-returned concentration scores S5.

RETURNED PERSISTENCE

EFFECT	RETURNED LANES	PERSISTENCE STATUS
Allocation	Historical · Expected · Simulated	Persistent
Selection	Historical · Expected · Simulated	Persistent negative
Interaction	Expected · Simulated	Persistent in two lanes

Server-returned persistence labels S5.

Analysis — Governed comparison fields returned

Observation: The server returned differences, concentration and persistence for three compatible lanes S5.
Why it matters: Cross-lane claims use governed fields. Uncertainty: One receipt limits time-series interpretation. What would change the view: A later comparison receipt returning different statuses.

07 Coverage and Limitations

All three lanes and the cross-lane comparison returned results

Historical	**Available**	Realized attribution returned S2
Ex ante	**Available**	Expected attribution returned S3
Simulation and comparison	**Available**	Scenario and cross-lane results returned S4 S5

A Appendix — Sources and methods

Governed source inventory

Based on 4 of 4 evidence areas.

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Capability gate	get_gradient_capabilities	2026-06-30	Passed
S2	Historical attribution	get_portfolio_attribution	2026-06-30	Passed
S3	Ex ante attribution	get_portfolio_ex_ante_attribution	2026-06-30	Passed
S4	Simulated attribution	get_gradient_capabilities	2026-06-30	Passed
S5	Cross-lane comparison	get_gradient_capabilities	2026-06-30	Passed
S6	Historical return context	get_portfolio_historical_returns	2026-06-30	Passed

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCE AUDIT TRAIL

TAG	KEY PARAMETERS	DIGEST
S1	illustrative sample	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S2	illustrative sample	sha256:bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb
S3	illustrative sample	sha256:cccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc
S4	illustrative sample	sha256:dddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd
S5	illustrative sample	sha256:eeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee
S6	fields=cumulative_growth; limit=100; illustrative sample	sha256:ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

RETURNED SIGN BY SEGMENT

LANE	SEGMENT	RETURNED EFFECT	RETURNED SIGN	SOURCE
Historical	Public Equity	+0.2pp	Positive	S2
Historical	Alternatives	-0.8pp	Negative	S2
Governed ex ante	Public Equity	+0.27pp	Positive	S3
Governed ex ante	Alternatives	+0.08pp	Positive	S3
Simulated	Public Equity	+0.3pp	Positive	S4
Simulated	Alternatives	-0.2pp	Negative	S4

Signs are returned or directly classified per lane; no local lane subtraction S2 S3 S4.

Method boundary

Each lane keeps its own method, period, benchmark role and currency, and values are shown as returned; cross-lane results appear only where the comparison tool returned them S1 S2 S3.